

DSA 8020 R Lab 3: Multiple Linear Regression II

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Contents

Percentage of Body Fat and Body Measurements	1
Load the dataset	1
Exploratory Data Analysis	2
Numerical summary	2
Graphical summary	3
General Linear F-Test	5
Prediction	7
Multicollinearity	7

Percentage of Body Fat and Body Measurements

Age, weight, height, and 10 body circumference measurements are recorded for 252 men. Each man's percentage of body fat was accurately estimated by an underwater weighing technique.

Data Source: Johnson R. *Journal of Statistics Education* v.4, n.1 (1996)

Load the dataset

Code:

```
library(faraway)
data(fat)
head(fat)
```

```
##   brozek siri density age weight height adipos  free neck chest abdom  hip
## 1   12.6 12.3  1.0708  23 154.25  67.75   23.7 134.9 36.2  93.1  85.2  94.5
## 2    6.9  6.1  1.0853  22 173.25  72.25   23.4 161.3 38.5  93.6  83.0  98.7
## 3   24.6 25.3  1.0414  22 154.00  66.25   24.7 116.0 34.0  95.8  87.9  99.2
## 4   10.9 10.4  1.0751  26 184.75  72.25   24.9 164.7 37.4 101.8  86.4 101.2
## 5   27.8 28.7  1.0340  24 184.25  71.25   25.6 133.1 34.4  97.3 100.0 101.9
## 6   20.6 20.9  1.0502  24 210.25  74.75   26.5 167.0 39.0 104.5  94.4 107.8
##   thigh knee ankle biceps forearm wrist
## 1  59.0 37.3  21.9   32.0   27.4  17.1
## 2  58.7 37.3  23.4   30.5   28.9  18.2
## 3  59.6 38.9  24.0   28.8   25.2  16.6
```

```
## 4  60.1 37.3 22.8  32.4   29.4 18.2
## 5  63.2 42.2 24.0  32.2   27.7 17.7
## 6  66.0 42.0 25.6  35.7   30.6 18.8
```

For the purposes of this lab, we will use only the following variables for conducting data analysis:

1. y **brozek**: Percent body fat using Brozek's equation

$$\frac{457}{\text{Density}} - 414.2$$

2. x_1 **age**: Age (yrs);
3. x_2 **weight**: Height (inches);
4. x_3 **height**: Height (inches);
5. x_4 **chest**: Chest circumference (cm);
6. x_5 **abdom**: Abdomen circumference (cm) at the umbilicus and level with the iliac crest

Code:

You can use the code below to extract these variables:

```
vars <- c("brozek", "age", "weight", "height", "chest", "abdom")
data <- fat[, vars]
```

Exploratory Data Analysis

Numerical summary

1. Use `summary` command to produce various numerical summaries of each of the 6 variables under consideration

Code:

```
summary(data)
```

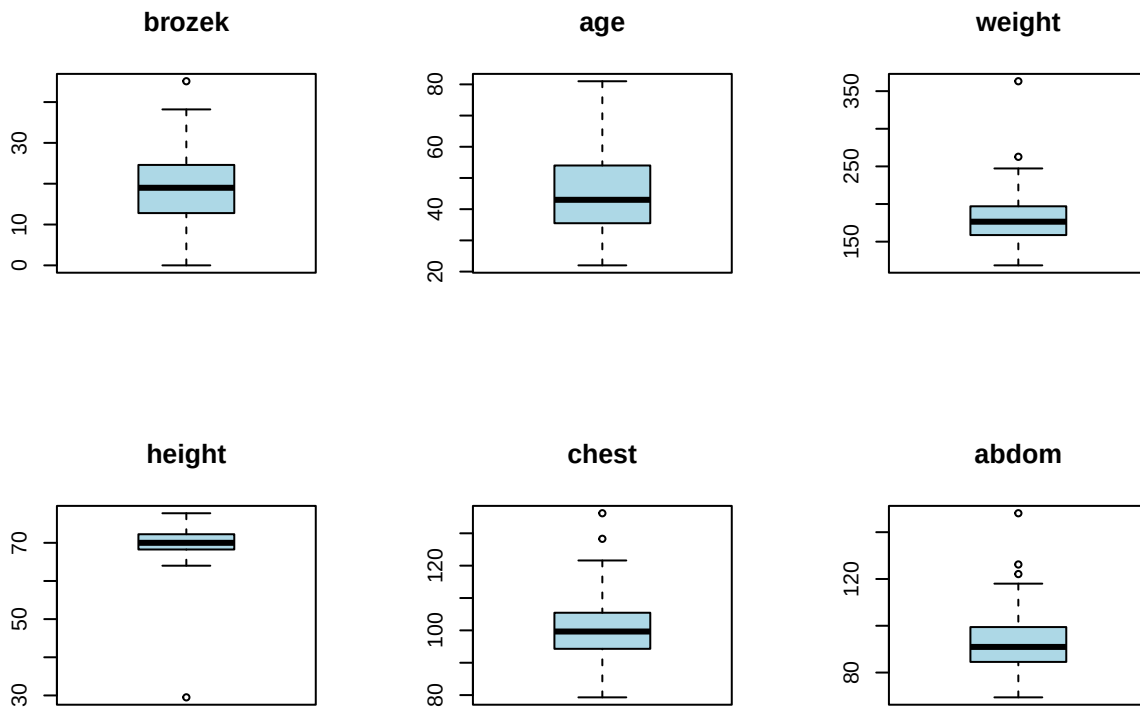
```
##      brozek      age      weight      height
##  Min.   : 0.00  Min.   :22.00  Min.   :118.5  Min.   :29.50
## 1st Qu.:12.80 1st Qu.:35.75 1st Qu.:159.0 1st Qu.:68.25
## Median :19.00 Median :43.00 Median :176.5 Median :70.00
## Mean   :18.94 Mean   :44.88 Mean   :178.9 Mean   :70.15
## 3rd Qu.:24.60 3rd Qu.:54.00 3rd Qu.:197.0 3rd Qu.:72.25
## Max.   :45.10 Max.   :81.00 Max.   :363.1 Max.   :77.75
##      chest      abdom
##  Min.   : 79.30  Min.   : 69.40
## 1st Qu.: 94.35 1st Qu.: 84.58
## Median : 99.65 Median : 90.95
## Mean   :100.82 Mean   : 92.56
## 3rd Qu.:105.38 3rd Qu.: 99.33
## Max.   :136.20 Max.   :148.10
```

Graphical summary

2. Make a boxplot for each variable

Code:

```
par(mfrow = c(2, 3))
for (var in vars) {
  boxplot(data[[var]], main = var, col = "lightblue")
}
```



3. Briefly discuss the shape of the distribution of each variable

Answer:

Brozek

- Slight right skew.
- Few high outliers on upper end.

Age

- Distribution appears symmetrical.

Weight

- Right-skewed distribution.
- Few high-value outliers.

Height

- Approximately symmetric.
- Single outlier on lower end.

Chest

- Right skew.
- Few high value outliers.

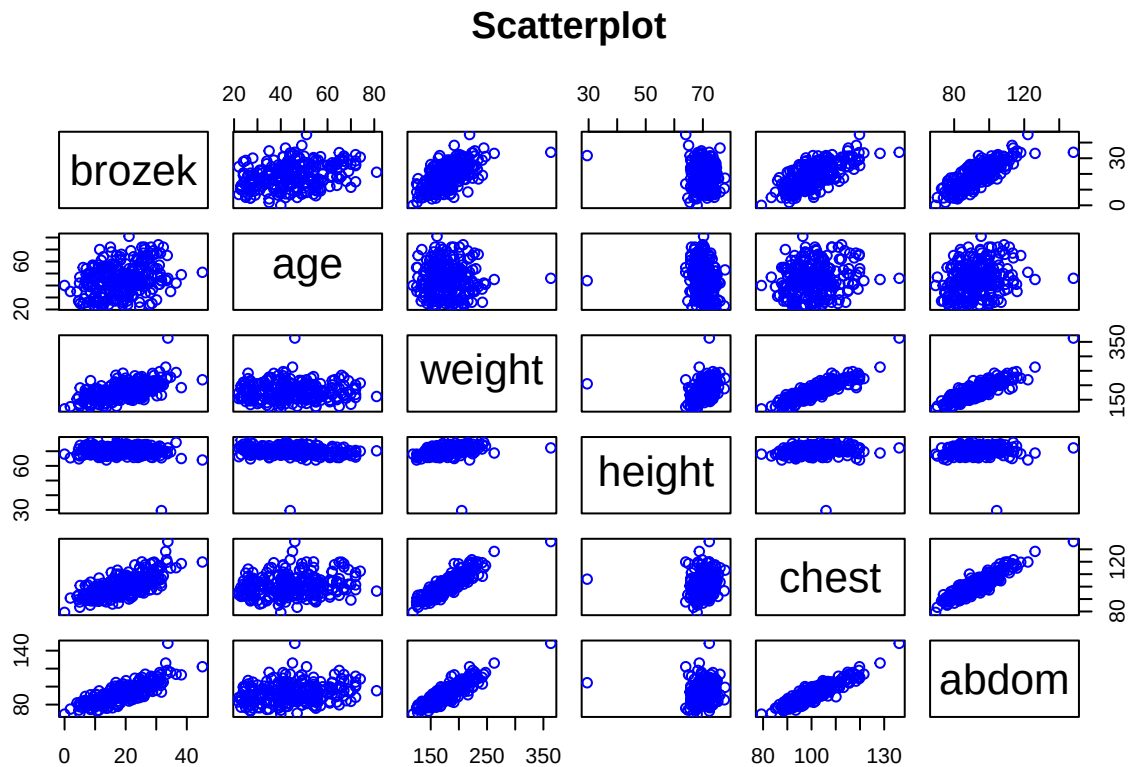
Abdomen

- Right skew.
- Multiple high value outliers.

4. Create a scatterplot matrix to explore the inter-dependence between these variables

Code:

```
pairs(data, main = "Scatterplot", col = "blue")
```



General Linear F-Test

Suppose a researcher would like to compare the “Full” model using all the 5 predictors and a “reduce” model where only x_1 (age) and x_5 (abdom) are used by performing a general linear F-test:

5. Write down the null and the alternative hypotheses

Answer:

Null Hypothesis (H_0)

The additional predictors do not improve the model, thus reduced model fits the data as well as the full.

Alt Hypothesis (H_1)

The full model greatly fits the data better than the reduced model, showing that the additional predictors are vital.

6. Fit the full model and write down the fitted linear regression equation.

Code:

```
full_model <- lm(brozek ~ age + weight + height + chest + abdom, data = data)
summary(full_model)
```

```
##
## Call:
## lm(formula = brozek ~ age + weight + height + chest + abdom,
##     data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11.6515  -2.9213   0.0552   2.9019   9.4269
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -32.153538    7.779978  -4.133 4.92e-05 ***
## age          -0.006447    0.024734  -0.261  0.795
## weight       -0.121843    0.028160  -4.327 2.20e-05 ***
## height       -0.118164    0.083492  -1.415  0.158
## chest        -0.012862    0.087484  -0.147  0.883
## abdom         0.894248    0.074150  12.060 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.134 on 246 degrees of freedom
## Multiple R-squared:  0.7212, Adjusted R-squared:  0.7155
## F-statistic: 127.2 on 5 and 246 DF,  p-value: < 2.2e-16
```

Answer:

$$\hat{brozek} = \beta_0 + \beta_1 * age + \beta_2 * weight + \beta_3 * height + \beta_4 * chest + \beta_5 * abdomen$$

7. Fit the reduced model and write down the fitted linear regression equation.

Code:

```
reduced_model <- lm(brozek ~ age + abdom, data = data)
summary(reduced_model)

##
## Call:
## lm(formula = brozek ~ age + abdom, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.7114  -3.2622   0.0285   3.2248  12.0577
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -36.51507     2.46972  -14.785  < 2e-16 ***
## age           0.06605     0.02290   2.884  0.00427 **
## abdom        0.56710     0.02677  21.187  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.45 on 249 degrees of freedom
## Multiple R-squared:  0.673, Adjusted R-squared:  0.6704
## F-statistic: 256.3 on 2 and 249 DF,  p-value: < 2.2e-16
```

Answer:

$$\hat{brozek} = \beta_0 + \beta_1 * age + \beta_2 * abdomen$$

8. Perform a general linear F-test and state the conclusion at $\alpha = 0.05$

Code:

```
anova(reduced_model, full_model)

## Analysis of Variance Table
##
## Model 1: brozek ~ age + abdom
## Model 2: brozek ~ age + weight + height + chest + abdom
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      249 4930.3
## 2      246 4204.7  3      725.6 14.151 1.543e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Answer:

We reject null hypothesis as p-value of 1.543e-08 much lower than our significance level $\alpha = 0.05$.

Prediction

9. Predict a future response for an individual with `age = 54`, `weight = 197`, `height = 72.25`, `chest = 105.375`, and `abdom = 99.325`. Construct a 95% prediction interval.

Code:

```
new_data <- data.frame(age = 54, weight = 197, height = 72.25, chest = 105.375, abdom = 99.325)
predict(full_model, newdata = new_data, interval = "prediction", level = 0.95)
```

```
##          fit          lwr          upr
## 1 22.42373 14.24419 30.60327
```

Answer:

For a patient with these characteristics, we can expect a body fat percentage of 22.4%.

10. Construct a 95% confidence interval for the mean response of percent body fat with `age = 54`, `weight = 197`, `height = 72.25`, `chest = 105.375`, and `abdom = 99.325`.

Code:

```
predict(full_model, newdata = new_data, interval = "confidence", level = 0.95)
```

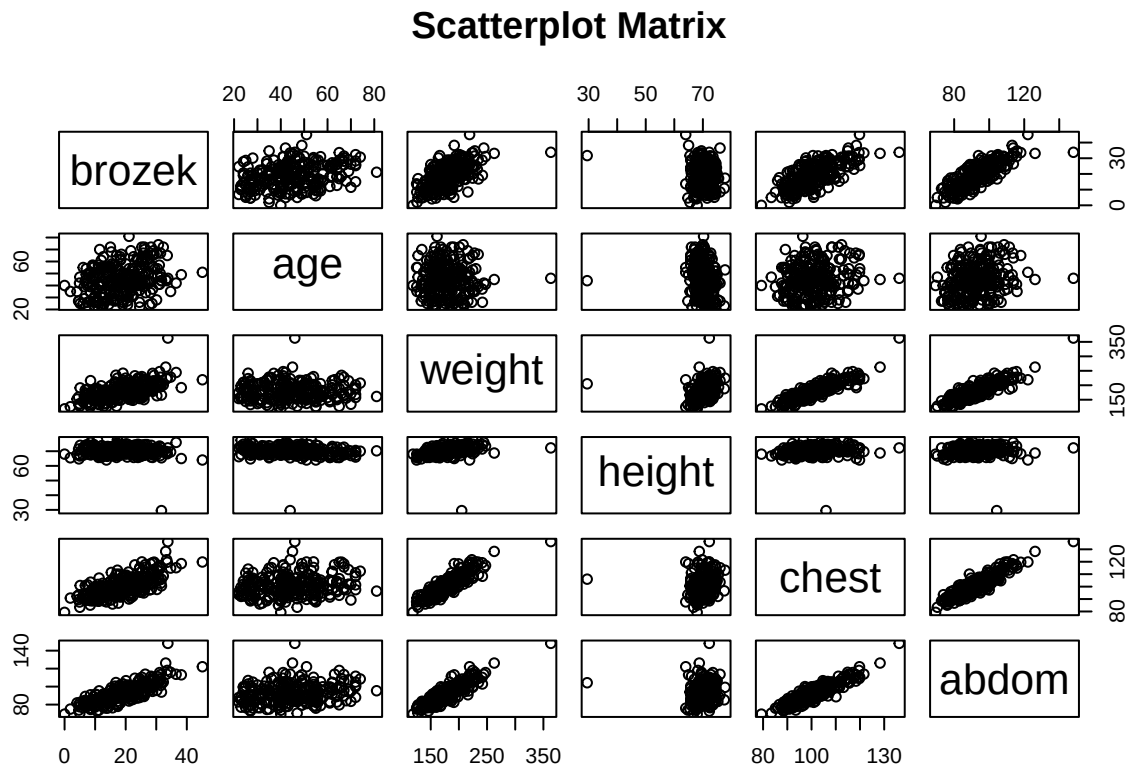
```
##          fit          lwr          upr
## 1 22.42373 21.65224 23.19523
```

Multicollinearity

11. Make the scatterplot matrix and compute the correlation matrix for all 6 variables (including the response).

Code:

```
pairs(data, main = "Scatterplot Matrix")
```



```
cor(data)
```

```
##          brozek      age      weight      height      chest      abdom
## brozek  1.0000000  0.28917352  0.61315611 -0.08910641  0.7028852  0.81370622
## age     0.28917352  1.00000000 -0.01274609 -0.17164514  0.1764497  0.23040942
## weight  0.61315611 -0.01274609  1.00000000  0.30827854  0.8941905  0.88799494
## height -0.08910641 -0.17164514  0.30827854  1.00000000  0.1348918  0.08781291
## chest   0.70288516  0.17644968  0.89419052  0.13489181  1.0000000  0.91582767
## abdom   0.81370622  0.23040942  0.88799494  0.08781291  0.9158277  1.00000000
```

12. Calculate VIF and briefly discuss your finding

Code:

```
vif(full_model)
```

```
##          age      weight      height      chest      abdom
##  1.426799 10.058282  1.373446  7.987963  9.388374
```

Answer:

No Multicollinearity:

- Age (1.42)
- Height (1.37)

Moderate Multicollinearity:

- Chest (7.98)

Severe Multicollinearity:

- Abdoment (9.38)
- Weight (10.05)